

The Significance of Asymmetrical Sexual Isolation

LEE EHRMAN

and

MARVIN WASSERMAN

(1987)

INTRODUCTION

Kaneshiro (1976) observed asymmetrical sexual isolation among the Hawaiian *Drosophila planitibia* subgroup of flies, and suggested a model of evolution of sexual behavior wherein females of putative ancestral species are sexually more isolated from males of derived species than are females of derived species from males of the ancestral species. The model has subsequently been expanded and new, allegedly supportive data presented (Kaneshiro, 1980, 1983; Ohta, 1978; Arita and Kaneshiro, 1979; Ahearn, 1980; Kaneshiro and Kurihara, 1981; Dodd and Powell, 1985). Ostensibly, the Kaneshiro model provides a reasonable explanation of asymmetrical sexual isolation, which may result from several underlying genetic mechanisms. Kaneshiro (1976) suggested that during a new invasion of a previously uninhabited locality, genetic drift acting upon a small number of migrants may cause the loss of several facets of male courtship behavior. Ancestral females would then be relatively unwilling to accept behaviorally deficient males. Derived females would accept these males as well as ancestral ones whose courtship pattern included

LEE EHRMAN • Division of Natural Sciences, State University of New York at Purchase, Purchase, New York 10577. MARVIN WASSERMAN • Biology Department, Queens College of the City University of New York, Flushing, New York 11367.

all that was still present in the derived males. Ohta (1978), studying yet another group of Hawaiian flies, showed that derived females had a higher sex drive and were more receptive to all males than were ancestral females. Perhaps in a small migrant population, where encounters between flies might be relatively rare events, those females that had a strong sex drive would have been selected for because they were more likely to mate than those females that were innately slower to mate. This scenario could lead to the loss of selection for the complete courtship pattern, particularly if there were no other closely related species present in the newer locality.

Kaneshiro (1980) felt that his model might have a general application and proposed that where present, asymmetrical isolation could indicate the direction of evolution. It is this claim (i.e., that the direction of asymmetrical isolation, when interpreted using the Kaneshiro model, accurately reflects the historical direction of evolution) that was criticized by Watanabe and Kawanishi (1979), Wasserman and Koepfer (1980), Markow (1981a), and Moodie (1982). They presented data and/or models wherein the direction of evolution was opposite to that proposed by Kaneshiro.

Taking these criticisms into consideration, Giddings and Templeton (1983) refined the Kaneshiro model, defining the specific conditions under which it would apply, and argued that, as redefined, the Kaneshiro model would reliably show the direction of evolution. They suggested that not only might drift lead to a simplification of the courtship pattern, but that founder events might "destabilize a primary stable mating system." If male courtship pattern changed, the females would, of necessity, accept these new changes without necessarily rejecting the old patterns. Derived females would therefore accept both old and new mating patterns, while ancestral females would accept only the old patterns. This scenario would lead to the same direction of evolution as that proposed by Kaneshiro; and Giddings and Templeton perhaps felt that they were simply clarifying his model. This redefinition of the Kaneshiro model, however, has not yet been accepted by Kaneshiro as a replacement for his original model (K. Y. Kaneshiro, personal communication, June 1985). Thus, we have two "Kaneshiro" models, the original Kaneshiro model and the Giddings and Templeton one.

We have no quarrel with these theoretical models; they are, after all, a few of the many alternative models of speciation that have been proposed (Bush *et al.*, 1977; Powell, 1982; Templeton, 1980; Carson and Templeton, 1984; Barton and Charlesworth, 1984). However, we do question the suggestion that given asymmetrical sexual isolation, one can with any confidence specify the direction of evolution between the two forms under investigation.

The question should not be whether the Kaneshiro model—or any other model—yields the correct direction of evolution under those conditions specified by the model, but rather: given asymmetrical isolation, can the direction of evolution be correctly determined? If asymmetrical isolation almost always arises for the same reason, e.g., as in the original Kaneshiro model, where the ancestral females are more isolated than the ancestral males from the derived species, one would have an accurate indication of the direction of evolution in those organisms where asymmetry exists. Asymmetry would therefore be an extremely important tool for studying and interpreting evolutionary processes. However, if any of the models is mistakenly assumed to be the exclusive method, or at least the most common method, by which asymmetrical isolation arises, its utilization as a valid predictive model would result in many incorrect conclusions. Unfortunately, sufficient data to test these models are lacking. We therefore wish to bring this controversial and crucial biological matter to the attention of those who may possess or be in the process of gathering the sort of data needed to test these models.

In order to evaluate a model, it is necessary to examine both underlying assumptions of the model and the design of the experiments used to test the model. The experiments usually employed are the so-called male choice tests, whereby one type of male is exposed to one or more of its own type of females and an equal number of foreign females. The isolation index I is usually calculated as the number of homogamic matings Ho minus the number of heterogamic matings He divided by the total number of matings N , where homogamic matings are between like individuals and heterogamic matings are between unlike individuals. Thus,

$$I = \frac{Ho - He}{N}$$

There is at least one assumption that is made by *all* of the models: namely that isolation indices are due solely to the behavior of the females, i.e., it is the females who control the amount and type of mating that takes place. Thus, any differences that may be observed are explained as being due to the behavior of the females.

Let us examine the implications of this assumption and the experimental design. We will assume that a total of 100 females of each type is exposed to 20 males of each type. Our first example is a simple one: the situation where there is no differential sex drive and/or discrimination, and tests using A males with A females and B females give exactly the same results as tests using B males with A females and B females.

	Number of females inseminated		
	A females	B females	Total
Test 1: A males	40	10	50
Test 2: B males	10	40	50
Total	50	50	100

The symmetrical results yield an isolation index I of 0.60 $[(40 - 10)/50]$ for both male choice tests. If one accepts the assumptions that the choice between mating and not mating is solely due to the females, and that the "male choice" tests are comparable, one can reasonably argue that female behavior between tests is comparable (by reading the columns). Thus, when exposed to 20 A males, 40 of 100 A females were willing to mate, while only 10 of 100 A females were willing to mate when exposed to B males. Again, the isolation indices are symmetrical and equal to 0.60.

There is some evidence that males also play an active role in the decision-making during courtship (Wasserman *et al.*, 1971; Markow, 1981*b*, 1982; Markow and Richmond, 1981). The more active the role played by males, the more likely it is that asymmetry will be observed and misinterpreted. Let us now suppose that although there is no change in the relative frequency of homogamic versus heterogamic matings, in test 2 below, the B males have a higher sex drive and mate twice as frequently as do the A males. Then our table looks as follows:

	Number of females inseminated		
	A females	B females	Total
Test 1: A males	40	10	50
Test 2: B males	20	80	100
Total	60	90	150

The results are no longer symmetrical. To calculate isolation indices *of females*, we read *down* the table and find that the index for the A females has dropped, while that of the B females has increased:

$$I_{A \text{ females}} = \frac{40 - 20}{60} = 0.33$$

$$I_{B \text{ females}} = \frac{80 - 10}{90} = 0.77$$

As a third case, let us suppose that both types of males are equally suc-

successful in mating with their own females, but that the B males are more discriminating than are the A males. Then our table might look as follows:

	Number of females inseminated		
	A females	B females	Total
Test 1: A males	40	10	50
Test 2: B males	5	45	50
Total	45	55	100

Under these conditions, the A female isolation index is higher than our original 0.60:

$$I_{A \text{ females}} = 0.77 \quad I_{B \text{ females}} = 0.60$$

Moreover, such results are no longer symmetrical. We also find that these tests do not allow us to distinguish between discriminating B males or discriminating A females.

Thus, differences in the behavior of the males will produce apparent differences in the isolation indices of the females when male choice tests are used. The type of male that has the higher sex drive and/or discriminates less than the other type of male will yield results that can be, and are, interpreted as a higher isolation index in favor of his own type of female. Without exception, all of the models discussed in this paper suffer from this difficulty.

Ohta's (1978) observation that the derived females had a higher sex drive than the ancestral females would not in itself produce an asymmetry:

	Number of females inseminated		
	A females	B females	Total
Test 1: A males	40	20	60
Test 2: B males	10	80	90
Total	50	100	150

$$I_{A \text{ females}} = 0.6 \quad I_{B \text{ females}} = 0.6$$

The fact that B females mate twice as often as do the A females does not change the isolation index. What is needed to accomplish such changes is an actual difference in the discriminatory ability of the females *or* any difference in the behavior of the males, or both.

We will now examine some of the models that yield asymmetrical isolation indices.

THE KANESHIRO MODELS

Although there are major differences between the original Kaneshiro model and that proposed by Giddings and Templeton, it is useful to discuss both models simultaneously. Kaneshiro has never placed any restrictions on his model. He apparently feels that given asymmetrical sexual isolation, the direction of evolution is as he proposes. The only real test of his model is to gather data and determine whether, in fact, given asymmetrical sexual isolation, the direction of evolution is as he proposes it should be in an overwhelming majority of cases.

Giddings and Templeton (1983), on the other hand, added two conditions to their concept of the Kaneshiro model:

1. "That these populations are related by founder events with little or no subsequent gene flow."
2. "That the ancestral populations have not been subjected to drastic population bottlenecks since these founder events."

As discussed above, to these conditions we add the proviso that:

3. All males behave in an identical manner.
4. Neither of the two populations can be subjected to selection for either sexual or nonsexual traits.
5. There must be no intermediate populations, extant or extinct, between the two populations in question.

The latter two constraints, although not specifically spelled out by Giddings and Templeton in their paper, were used to clarify their concept of Kaneshiro's model and to explain results where the direction of evolution did not fit the Kaneshiro model. We now discuss some of these examples.

Selection for Nonsexual Behavioral Traits

Markow (1981a), using *Drosophila melanogaster*, selected for positive and negative geotropism and phototropism. She found that the derived populations showed asymmetrical isolation in male choice tests with the original ancestral base population. However, the direction of the asymmetry often did not conform to that predicted by the Kaneshiro model. Giddings and Templeton (1983) discount these experimental results, stating that her experiment was not a good test of the Kaneshiro model. As they interpreted it, Markow was selecting for nonsexual dif-

ferentiation and therefore her experiment was an example of "speciation via adaptive divergence" rather than speciation by a founder event [terminology of Templeton (1980)]. However, although Markow's data do not test the Giddings and Templeton model, they do bear on the Kaneshiro model in that Markow demonstrates a means by which asymmetrical isolation may arise—an asymmetrical isolation which, when interpreted by the Kaneshiro model, yields the "wrong" direction of evolution.

Evidence from the Hawaiian *Drosophila*

We feel that some of the critical data upon which the original Kaneshiro model is based permit more than one interpretation. For example, the researchers working on the Hawaiian *Drosophila* species usually assume that the older species of *Drosophila* are found on the older islands, while the younger species arose by founder events as flies invaded newly formed, younger islands. Thus, Kaneshiro (1976) assumed that within the *D. planitibia* subgroup, *D. differens* (D) which resides on the oldest island in question, Molokai, is the oldest species; *D. planitibia* (P) which lives on Maui is next in age; while *D. heteroneura* (H) and *D. silvestris* (S), living on the youngest and largest Hawaiian island, the island of Hawaii, are the youngest species (see Carson and Bryant, 1979). Kaneshiro's model of asymmetry indicated that the direction of evolution was from $D \rightarrow P \rightarrow S$; with H coming from P, or, more likely, directly from D. (Parenthetically, his model yields two possible ancestors for H—an unsatisfactory situation, which can lead not only to theoretical problems, but also to practical difficulties. This will be discussed further below in the section on the *D. mojavensis*-*D. arizonensis* case.) Moreover, according to Kaneshiro, S and H resulted from two separate founder events, having as their immediate ancestor either P or D.

Recently published data cause us to question these conclusions. Using chromosomal sequences to indicate evolutionary direction in his review of the Hawaiian *Drosophila*, Carson (1983) showed that migration of flies was not always from older islands to newer ones. One of the six founder events between Kauai and the younger island of Oahu was in the "wrong" direction; eight of the 18 events between the Maui complex and the older islands were also "wrong". The youngest island, Hawaii, is believed to have received 19 invasions, but has contributed nothing to the older islands. Thus, 34 of 43 (almost 80%) of the invasions were in the "correct" direction; while 20% of the invasions were in the "wrong" direction. Perhaps one should not be unhappy with an overall batting

average of 0.800, unless, of course, the species one is studying happen to be among the 20%.

In the *D. planitibia* subgroup, biochemical data reviewed by Hunt and Carson (1983) are both available and informative. Both allozyme studies and DNA hybridization work indicate that a single ancestor gave rise to two lines, one leading to P and D and the other leading to H and S. According to their data, species H and S were therefore not independently derived from P and D by two separate invasions of Hawaii. Only one invasion of Hawaii occurred; H and S were derived from that single migrant that probably came from a population that was also ancestral to the P and D line. The speciation separating H and S did not involve interisland movement, but occurred within the island of Hawaii. The fact that interisland migration is not a necessary prerequisite for speciation has been demonstrated by Carson and Bryant (1979) and Carson *et al.* (1982), who presented morphometric evidence to show that incipient speciation was taking place within the species S on Hawaii. Other evidence to support the close relationship between S and H are the following facts:

1. Both are polymorphic for a unique inversion not known to exist elsewhere.
2. Under laboratory conditions, S and H will mate with each other and produce fully fertile offspring, while all other combinations of crosses between either one of these species with other members of the *D. planitibia* subgroup produce sterile males.
3. Kaneshiro and Val (1977) have evidence that a considerable amount of hybridization is taking place between H and S on Hawaii.

Using Giddings and Templeton's first condition ("little or no subsequent gene flow"), this latter situation negates the *D. planitibia* group as a good example of the Kaneshiro model as they interpret it. We also note that according to Giddings and Templeton (1983), "Founder effects in *Drosophila* of the type that lead to reproductive isolation tend preferentially to lose inversion polymorphisms." Yet, according to the Kaneshiro proposal, the two monomorphic species P and D gave rise to the polymorphic species S and H by two separate founder events. We are forced to conclude that the *D. planitibia* subgroup, the example for which the model was originally proposed, certainly is not a good example of the Giddings and Templeton model and is most likely not even a good example of the original Kaneshiro model.

The work of Ohta (1978) on *D. grimshawi* is often cited as contributing supporting evidence for the Kaneshiro model. However, Ohta's data show that the model did not work for the *D. grimshawi* flies on the Maui com-

plex of islands, perhaps because these islands are close to each other and there had probably been a series of founder events and subsequent gene flow among these islands. Giddings and Templeton call this the "solitary exception." However, if the model is unreliable as regards to the *D. grimshawi* flies on the Maui complex of islands, it may also be unreliable for the *D. planitibia* flies on these same islands. We therefore see no reason to prefer Kaneshiro's suggestion that *D. differens* (D) gave rise to *D. planitibia* (P) over that of Hunt and Carson (1983), who stated that *D. differens* and *D. planitibia* are daughter species sharing a common ancestor.

Other Evidence

Giddings and Templeton (1983) suggested that the work of McPhail (1969) on *Gasterosteus* supports the model, since the asymmetry is in the correct direction. However, they state that "an isolated population . . . has, in response to predation pressure, differentiated in pigmentation of the nuptial pouch" (our italics). Clearly these data, like Markow's, result from "speciation via adaptive divergence" and are therefore not a good example of the Kaneshiro model. If anything, the *Gasterosteus* example is worse than Markow's, since it is a case of selection for a character directly involved in mating behavior. Recall that in the Markow experiments, she was selecting for nonsexual behavioral traits. Any sexual isolation that arose in her experiments could be attributed to drift acting on small population samples.

Some of the other examples used as evidence that the Kaneshiro model is a common phenomenon are not totally convincing. Thus, Giddings and Templeton claim that *D. subobscura* fits the Kaneshiro model because the populations from Central Europe discriminate against those from Norway (Sperlich, 1964). It is not at all certain, however, that the *D. subobscura* populations of Norway arose "by founder events with little or no subsequent gene flow," the first of the two essential assumptions that they specify.

THE WATANABE-KAWANISHI MODELS

Watanabe and Kawanishi (1979, 1981, 1983) recorded asymmetrical sexual isolation in both the *D. melanogaster* and the *D. virilis* species groups. They agreed with Kaneshiro in assuming that one can determine

the direction of evolution by the asymmetry in sexual isolation, but concluded that the direction should be exactly opposite that proposed by Kaneshiro. Their first model (Watanabe and Kawanishi, 1979, 1981) was discussed by Wasserman and Koepfer (1980), who pointed out that it described a sympatric speciation process. According to Watanabe and Kawanishi, the females of the newer incipient species must quickly learn to discriminate against the males of the ancestral species or else the newer species would never be able to fully diverge. On the other hand, mating between the males of the new species and the females of the ancestral species would not prevent the new species from diverging if the newer forms are relatively few in number. The development of this asymmetrical isolation can only occur when the individuals are in direct contact with each other; hence, this is a sympatric speciation model and sympatric speciation among forms that can only reproduce by outcrossing is believed by most zoologists to be extremely rare, if it ever occurs at all. Perhaps for this reason this Watanabe-Kawanishi model has not generated much support.

Later Watanabe and Kawanishi (1983) presented data for a second model, one that they considered as being due to the stasipatric speciation of White (1968). Here again asymmetry in sexual isolation was observed. The direction of evolution was the same as that derived from their sympatric model, but opposite to that of Kaneshiro's model. Their utilization of the stasipatric model is not convincing. Two requirements of this model are the presence of a chromosomal rearrangement that yields negative heterosis and an extremely sedentary organism. White's grasshoppers fulfill both of these requirements; the *Drosophila* of Watanabe and Kawanishi fulfill neither.

It is of some interest to note that Robertson (1983) confirmed that the direction of evolution of *D. mauritiana* from *D. simulans* follows that of the Watanabe-Kawanishi model rather than that of the Kaneshiro model. Robertson, however, rejected their sympatric model and the character displacement model of Wasserman and Koepfer (1980), pointing out that *D. mauritiana* most probably arose when geographically isolated from *D. simulans*. He concluded that the asymmetry between these species is probably unrelated to the direction of evolution.

OTHER MODELS

Evidence from *Drosophila arizonensis*-*Drosophila mojavensis* Case

Wasserman and Koepfer (1977) noted character displacement for sexual isolation between *D. mojavensis* and *D. arizonensis*. T. K. Watanabe

and M. Kawanishi (personal communication) pointed out that the data showed asymmetrical isolation. This case was reviewed by Wasserman and Koepfer (1980) and can be briefly summarized here.

Cytological, morphological, behavioral, and biochemical evidence all show that *D. arizonensis* and *D. mojavensis* are sibling species; there is no evidence for any intermediate forms [reviewed by Heed (1978)]. In their studies, Wasserman and Koepfer (1977, 1980) were concerned with three populations: a *D. mojavensis* population from Baja California, which is separated by the Gulf of California from both a second population of *D. mojavensis* and the species *D. arizonensis*, both of which are sympatric in Sonora. The proposed speciation scenario was as follows: The ancestral species was divided by the Gulf of California. Then, the two isolated populations diverged ecologically, behaviorally, and to a certain degree reproductively. At least some of the resulting behavioral differentiation could be attributed to the fact that the Baja flies are not sympatric with any close relatives with which they might conceivably mate, while the mainland Sonoran flies are sympatric with close relatives and therefore had to develop and/or maintain their mate recognition abilities. Migrants from Baja invaded Sonora. These were morphologically, ecologically, and behaviorally different from endemic Sonoran flies. An either/or situation then existed: Either the founders would mate with the local flies and lose their distinctiveness, or selection would increase the isolation, amplifying their differences. In this case, the isolation was reinforced, with the small migrant population changing more than the larger endemic form. Our results agree with King (1985), who suggested that in a situation similar to that found here, the smaller population would evolve more quickly than the larger one. The reinforcement of the sexual isolation proceeded to the point where the speciation process was completed, resulting in character displacement for sexual isolation between the endemic Sonoran form, now called *D. arizonensis*, and the species, *D. mojavensis*, which consists of the founder Sonoran population plus its parental Baja population. A third population in California, which was not involved in this process and is a northern branch of the Baja flies, is also designated as *D. mojavensis*.

As a by-product of these speciation events, both intra- and interspecific asymmetrical isolation exists under laboratory conditions. Baja *D. mojavensis* females are relatively undiscriminating and will readily accept Sonoran *D. mojavensis* males and, to a lesser extent, *D. arizonensis* males. Sonoran *D. mojavensis* females tend to reject *D. arizonensis* males and show a small, but significant isolation from Baja *D. mojavensis* males. The *D. arizonensis* females are strongly isolated sexually from both populations of *D. mojavensis* males. The resulting direction of asymmetry is actually in the reverse direction from that suggested by the Kaneshiro

model. Although the direction is similar to that proposed by Watanabe and Kawanishi, neither sympatric speciation nor stasipatric speciation was involved in the evolution of these flies. The reasons for the asymmetry observed between the various populations of *D. arizonensis* and *D. mojavensis* are as follows:

1. Asymmetry between Baja *D. mojavensis* and *D. arizonensis*. The Baja flies occur in a region where there are no other close relatives. These flies do not have to maintain a highly discriminatory posture. The *D. arizonensis* flies are exposed to several other *D. mulleri* complex species. They are therefore, by necessity, more discriminating.

2. Asymmetry between Baja *D. mojavensis* and Sonoran *D. mojavensis*. Selection in Sonora led to a greater ability in Sonoran *D. mojavensis* for capacity to discriminate against *D. arizonensis*. By so doing, these flies apparently learned to discriminate not only against *D. arizonensis*, but also against other "foreign" flies. This type of behavior was observed by Koepfer (1984), who selected for increased isolation between Sonoran and Baja *D. mojavensis* flies. She reported that those selected lines that exhibited an increased isolation between the Sonoran and the Baja flies also showed an increased isolation from California *D. mojavensis*, which represented "foreign" flies to which her selection lines had not been historically exposed.

If one were to assume that the Kaneshiro model was in operation here, one would not only conclude that Baja *D. mojavensis* is the most derived form, but also that it could have had two ancestors—both Sonoran *D. mojavensis* and *D. arizonensis*. This could be interpreted to mean that either false-positive results are possible, or perhaps Baja *D. mojavensis* was derived from the mainland prior to the speciation event that separated *D. mojavensis* from *D. arizonensis* there. However, if one accepts this scenario, then there is no longer a mechanism to explain the speciation between the two sympatric mainland forms. In any case, had only one of the two Sonoran populations been studied along with the Baja population and interpreted according to the Kaneshiro model, the data would have indicated that that particular Sonoran population (whichever one happened to be chosen by the investigator) is the ancestor of the Baja population, and that the migration proceeded from Sonora to Baja. The fact that the Kaneshiro model can yield two or more possible ancestors should give pause to investigators who might wish to use this method. Obtaining a single positive result is not sufficient—all potential populations must still be tested.

Giddings and Templeton (1983) stated that the *D. arizonensis*-*D. mojavensis* case (Wasserman and Koepfer, 1977, 1980) was not a good example of the Kaneshiro model—as they define it—because of Was-

serman's (1954, 1962) suggestion that many secondary hybridizations could account for the cytological evolution that has occurred in the *Drosophila mulleri* complex (their initial condition requiring no gene flow). However, multiple hybridizations as a mechanism for evolution in the *D. mulleri* complex was completely rejected in those papers and has been discounted in all other publications where the evolution of these forms was discussed [see review in Wasserman (1982)]. Although we disagree with their reasons, we do agree with Giddings and Templeton in their conclusion that the *D. arizonensis*-*D. mojavensis* situation is not a valid test of the Kaneshiro model as they interpret it. However, it certainly has a bearing on the reliability of putative correlations between the direction of evolution and the direction of asymmetrical sexual isolation. This is another example where an experimenter, finding asymmetrical sexual isolation and accepting the Kaneshiro model as a general rule, would arrive at entirely incorrect conclusions.

Evidence from *Drosophila paulistorum*

Most of the processes of race divergence and of formation of isolating mechanisms take place while populations are allopatric or only slightly overlapping in their geographical distributions. Reproductively isolated populations may continue to be more or less strictly allopatric. Perhaps more usually their distribution regions will overlap, making them sympatric in more or less extensive territories.

Natural selection favors situations that minimize or eliminate the gene exchange between divergent, and diverging, populations. These selection processes involve the appearance of rarely a single and more commonly a group of several reproductive isolating mechanisms. Very often none of these isolating barriers alone is sufficient to make the isolation complete, but they do so in combination. It should be noted that sympatric coexistence does not necessarily require the reproductive isolation to be absolute; it must, however, be sufficiently strong to make the gene exchange between the species populations not exceed the rate at which the diffusing genes can be eliminated by natural selection.

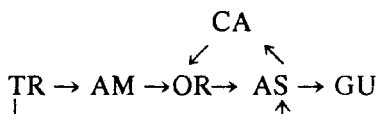
The *Drosophila paulistorum* complex seems to reside between the two stages outlined above. *Drosophila paulistorum* is not a single species, but rather a species complex consisting of six semispecies or incipient species. These species do not interbreed in nature. They are partially separated by their geographical distribution. In addition, there are three intrinsic isolating mechanisms: sexual isolation, hybrid sterility, and hybrid inviability. When produced in the laboratory, the female hybrids are

fertile, but the male hybrids are sterile due to the presence of endosymbiotic microbes [reviewed by Somerson *et al.* (1984a,b)].

The most effective isolating mechanism is the behavioral one, seen in those few cases where the geographical distributions of two or more semispecies overlap. There is evidence that natural selection has reinforced reproductive isolation in the wild, sympatric *D. paulistorum* populations. Direct observations in multiple choice experiments yielded average isolation coefficients of +0.67 in allopatric crosses and +0.85 in sympatric crosses (Ehrman, 1961). This is clearly an example of character displacement for sexual isolation.

The genetic basis of the sexual isolation was studied by examining the effects of individual chromosomes in hybrid flies (Ehrman, 1961). It was found that sexual preference is decided by the semispecies that contributes more than half of the genome. No one chromosome is more important than the others; the mating choice is controlled by genes on each of the three pairs of chromosomes.

Male choice tests have been run by Carmody *et al.* (1962). The six semispecies then identified were tested and data were obtained for all 15 possible combinations. We have reorganized these data, examining them for asymmetrical female isolation (Table I). The crosses between Transitional and the (then) Guianan data could not be analyzed because reciprocal crosses were not done. The data were pooled and the indices treated as binomials. Normal deviates were calculated and *p* values determined. Seven of the 14 pairs showed significant asymmetrical isolation. Using the Kaneshiro model, one would arrive at the following phylogeny:



The seven arrows represent those seven statistically significant asymmetries and give the proposed directions of evolution according to the Kaneshiro model. For the Watanabe-Kawanishi model(s), one would reverse the directions of the arrows. Where there are no connecting arrows, there was no significant asymmetry. Thus, for example, no significant asymmetry was observed between TR and OR, and between TR and CA; recall that the TR × GU crosses could not be analyzed.

There are, however, several problems with both the Kaneshiro phylogeny and the reverse Watanabe-Kawanishi phylogeny. First, both suffer from having an obvious circularity. According to Kaneshiro (Watanabe-Kawanishi), OR is the ancestor (descendant) of AS, which is the

TABLE I. Asymmetrical Isolation Indices for *Drosophila paulistorum* Semispecies^a

Females and homogamic males	Heterogamic males	Number of females	<i>I</i>	Asymmetry <i>p</i>
CA	TR	449	0.630	
TR	CA	441	0.628	n.s.
CA	AM	138	1.000	
AM	CA	134	1.000	n.s.
CA	AS	288	0.606	
AS	CA	282	0.845	0.001
CA	OR	139	0.952	
OR	CA	140	0.751	0.001
CA	GU	36	1.000	
GU	CA	39	1.000	n.s.
AM	TR	511	0.494	
TR	AM	461	0.579	0.008
AM	AS	123	0.896	
AS	AM	124	0.913	n.s.
AM	OR	330	0.960	
OR	AM	335	0.798	0.001
AM	GU	100	0.810	
GU	AM	96	0.859	n.s.
OR	TR	933	0.756	
TR	OR	940	0.784	n.s.
OR	AS	348	0.894	
AS	OR	326	0.701	0.001
OR	GU	260	0.842	
GU	OR	265	0.898	n.s.
AS	TR	788	0.255	
TR	AS	780	0.638	0.001
AS	GU	100	0.952	
GU	AS	97	0.829	0.006

^a CA, Centro-American; TR, Transitional; OR, Orinocan; AM, Amazonian; AS, Andean-South Brazilian; GU, Guianan. There are now more recently described Interior semispecies in Ehrman and Powell (1982).

TABLE II. Average Sexual Isolation Indices
between *Drosophila paulistorum*
Semispecies^a

Semispecies	Females	Males
CA	0.830	0.846
TR	0.654	0.514
AM	0.840	0.826
AS	0.709	0.722
OR	0.804	0.863
GU	0.896	0.909

^a See Table I for abbreviations.

ancestor (descendant) of CA, which in turn is the ancestor (descendant) of OR. Other examples of circularity are not as obvious: How does one interpret the "fact" that TR gave rise to AM, which led to OR (OR gave rise to AM, which led to TR); yet TR and OR are not statistically significantly different from each other? In fact, each of the seven nonsignificant *p* values results in at least one such circularity.

A second major discrepancy exists between both types of models and the *D. paulistorum* data. The Kaneshiro models assume that the ancestral females are more discriminatory than are the derived females. Table II gives the average isolation indices of both males and females of each of the five semispecies. One finds that the females of the Transitional semispecies, the presumptive ancestor of the complex, have the *lowest* average isolation index, not the highest. Whence comes the asymmetry? The Transitional males have the lowest average isolation index when compared to the other males (Table II). The reason for the asymmetry is, therefore, not because the TR females are more discriminatory than the other females—they are actually less discriminating. Rather, this asymmetry is due to the relatively greater mating success of the Transitional males in crossing with foreign females (as described in the second example above, p. 4).

A similar "Catch-22" exists for the Watanabe-Kawanishi models. According to them, the asymmetry indicates that TR is the most derived form and therefore should be sexually the most isolated from the others. Yet it is the least isolated semispecies.

One might argue that the Transitional semispecies probably is the modern representative of the ancestral population, a relict. However, the strongest evidence for this would be the fact that it is sexually *less* isolated from the other forms, rather than more isolated. Moreover, at least one

strain of TR produces fertile F_1 males when crossed to other semispecies. All other intersemispecies crosses produce only sterile males.

From all the manifestly comprehensive *D. paulistorum* data, merely hinted at above, with its chromosomal controls and crucial microbiology, it is still impossible to unequivocally determine the direction of evolution of these species. The evidence it provides neither supports nor refutes the Kaneshiro model. Why? Perhaps because the superspecies *D. paulistorum* is a mainland form flourishing on the continent of South America, and for this species complex, as for other continentally distributed forms, there probably have been no discrete founder events (Ehrman and Powell, 1982; Daniels *et al.*, 1984).

DISCUSSION

The question we are here addressing is: Given the presence of asymmetrical isolation, can one determine, with any degree of confidence and accuracy, the direction of evolution? We suggest that the answer to this question would be yes if the Kaneshiro model, or the Watanabe-Kawanishi model(s), or indeed any model, were the only means or at least the most common means by which asymmetrical isolation arises. This unfortunately is not likely to be the case.

It is generally accepted that two populations that are geographically isolated from each other will in time diverge. Often reproductive isolation arises as an accidental by-product of genetic divergence. As populations become more adapted to different environments, they become different in progressively more and more genes. Reproductive isolation may arise because the action of many genes is pleiotropic. Some gene differences selected for different reasons or resulting from random genetic drift may thus have isolating side effects. Alternatively, selection may directly act to build and reinforce the barriers to gene exchange between populations whose hybridization results in reproductive wastage (Ehrman and Parsons, 1981). It would therefore truly be remarkable if all allopatric populations developed their sexual isolation in exactly the same way and at the same rate. During the period when sexual isolation is not complete, asymmetrical isolation is probably the rule; however, its detection may be contingent upon the relative tenacity of the investigator. (For example, to obtain significant results when there is a low level of asymmetry may require many more tests than most investigators are willing to carry out.) Its "direction" would be dependent upon the many factors in operation,

including relative time to sexual maturity and discrimination and relative sex drive of the males.

We feel that the models discussed here are only a few of a great number of possible speciation scenarios which might include a temporary stage involving asymmetrical isolation. In many situations, the direction of evolution would be similar to that proposed by Kaneshiro, in others similar to that proposed by Watanabe and Kawanishi. The direction of asymmetrical isolation, taken by itself, is an unreliable indicator of the direction of evolution. We must admit, finally, that such theoretical arguments should carry little weight when compared to data. How could such data be acquired?

One must accumulate good examples where asymmetrical sexual isolation is present *and* where other evidence points to the correct direction of evolution. Then, one can determine which of the two contrasting models fits each of the examples. Once enough data have been accumulated, it might be possible to answer the question: Given the presence of asymmetrical isolation, can one determine, with any degree of confidence and accuracy, the direction of evolution?

We therefore request that those investigators who have published or unpublished data on asymmetrical isolation where the direction of evolution is known because of other factors publicize their information, indicating whether or not their example fits any of the models. (We would be delighted to receive such evidence for the purpose of compilation and comparison.)

REFERENCES

- Ahearn, J. N., 1980, Evolution of behavioral isolation in a laboratory stock of *Drosophila silvestris*, *Experientia* **36**:63-64.
- Arita, L. H., and Kaneshiro, K. Y., 1979, Ethological isolation between two stocks of *Drosophila adianta* Hardy, *Proc. Hawaii. Entomol. Soc.* **13**:31-34.
- Bush, G., Case, S., Wilson, A., and Patton, J., 1977, Rapid speciation and chromosomal evolution in mammals, *Proc. Natl. Acad. Sci. USA* **74**:3942-3946.
- Barton, N. H., and Charlesworth, B., 1984, Genetic revolutions, founder effects, and speciation, *Annu. Rev. Ecol. Syst.* **15**:133-164.
- Carmody, G., Collazo, A. D., Dobzhansky, T., Ehrman, L., Jaffrey, I. S., Kimball, S., Obrebski, S., Silagi, S., Tidwell, T., and Ullrich, R., 1962, Mating preferences and sexual isolation within and between the incipient species of *Drosophila paulistorum*, *Am. Midl. Nat.* **68**:67-82.
- Carson, H. L., 1983, Chromosomal sequences and interisland colonizations in Hawaiian *Drosophila*, *Genetics* **103**:465-482.
- Carson, H. L., and Bryant, P. J., 1979, Change in the secondary sexual character as evidence

- of incipient speciation in *Drosophila silvestris*, *Proc. Natl. Acad. Sci. USA* **76**:1929–1932.
- Carson, H. L., and Templeton, A. R., 1984, Genetic revolutions in relation to speciation phenomena: The founding of new populations, *Annu. Rev. Ecol. Syst.* **15**:97–131.
- Carson, H. L., Val, F. C., Simon, C. M., and Archie, J. W., 1982, Morphometric evidence for incipient speciation in *Drosophila silvestris*, *Evolution* **36**:132–140.
- Daniels, S. B., Strasbaugh, L. D., Ehrman, L., and Armstrong, R., 1984, Sequences homologous to *P* elements occur in *Drosophila paulistorum*, *Proc. Natl. Acad. Sci. USA* **81**:6794–6797.
- Dodd, D. M. B., and Powell, J. R., 1985, Founder–flush speciation: An update of experimental results with *Drosophila*, *Evolution* **39**:1388–1392.
- Ehrman, L., 1961, The genetics of sexual isolation in *Drosophila paulistorum*, *Genetics* **46**:1025–1038.
- Ehrman, L., and Parsons, P. A., 1981, *Behavior Genetics and Evolution*, McGraw-Hill, New York.
- Ehrman, L., and Powell, J. R., 1982, The *Drosophila willistoni* species group, in: *The Genetics and Biology of Drosophila*, Vol. 3b (M. Ashburner, H. L. Carson, and J. N. Thompson, Jr., eds.), pp. 193–225.
- Giddings, L. V., and Templeton, A. R., 1983, Behavioral phylogenies and the direction of evolution, *Science* **220**:372–378.
- Heed, W. B., 1978, Ecology and genetics of desert *Drosophila*, in: *Ecological Genetics: The Interface* (P. F. Brussard, ed.), pp. 109–126, Springer-Verlag, Heidelberg.
- Hunt, J. A., and Carson, H. L., 1983, Evolutionary relationships of four species of Hawaiian *Drosophila* as measured by DNA reassociation, *Genetics* **104**:353–364.
- Kaneshiro, K. Y., 1976, Ethological isolation and phylogeny in the *planitibia* subgroup of Hawaiian *Drosophila*, *Evolution* **30**:740–745.
- Kaneshiro, K. Y., 1980, Sexual isolation, speciation and the direction of evolution, *Evolution* **34**:437–444.
- Kaneshiro, K. Y., 1983, Sexual selection and the direction of evolution in the biosystematics of Hawaiian *Drosophila*, *Annu. Rev. Entomol.* **28**:161–178.
- Kaneshiro, K. Y., and Kurihara, J. S., 1981, Sequential differentiation of sexual behavior among populations of *Drosophila silvestris*, *Pac. Sci.* **35**:177–183.
- Kaneshiro, K. Y., and Val, F. C., 1977, Natural hybridization between a sympatric pair of Hawaiian *Drosophila*, *Am. Nat.* **111**:897–902.
- King, J. L., 1985, Selection for mating barriers between populations of unequal size, *Evol. Theory* **7**:149–151.
- Koepfer, H. R., 1984, Selection for increased sexual isolation between geographic forms of *Drosophila mojavensis*, Ph. D. thesis, City University of New York.
- Markow, T. A., 1981a, Mating preference is not predictive of the direction of evolution in experimental populations of *Drosophila*, *Science* **213**:1405–1407.
- Markow, T. A., 1981b, Courtship behavior and the control of reproductive isolation between *Drosophila mojavensis* and *Drosophila arizonensis*, *Evolution* **35**:1022–1026.
- Markow, T. A., 1982, Mating systems in cactophilic *Drosophila*, in: *Ecological Genetics and Evolution: The Cactus–Yeast–Drosophila Model System* (J. S. F. Barker and W. T. Starmer, eds.), pp. 273–287, Academic Press, New York.
- Markow, T. A., and Richmond, R. C., 1981, *Drosophila* males preferentially court virgin females, Eleventh Annual Meeting of Behavior Genetics Association, Abstract.
- McPhail, J., 1969, Predation and the evolution of a stickleback (*Gasterosteus*), *J. Fish. Res. Board Can.* **26**:3183–3208.
- Moodie, G. E. E., 1982, Why asymmetric mating preference may not show the direction of evolution, *Evolution* **36**:1096–1097.

- Ohta, A. T., 1978, Ethological isolation and phylogeny in the *grimshawi* species complex of Hawaiian *Drosophila*, *Evolution* 32:485-492.
- Powell, J. R., 1982, Genetic and non-genetic mechanisms of speciation, in: *Mechanisms of Speciation* (C. Barigozzi, ed.), pp. 67-74, A. R. Liss, New York.
- Robertson, H. M., 1983, Mating behavior and the evolution of *Drosophila mauritania*, *Evolution* 37:1283-1293.
- Somerson, N., Ehrman, L., and Kocka, J., 1984a, Induction by streptococcal L-forms of two reproductive isolating mechanisms in *Drosophila paulistorum*, *Isr. J. Med. Sci.* 20:1009-1012.
- Somerson, N., Ehrman, L., Kocka, J., and Gottlieb, F., 1984b, Streptococcal L-forms isolated from *Drosophila paulistorum* semispecies cause sterility in male progeny, *Proc. Natl. Acad. Sci. USA* 81:282-285.
- Sperlich, D., 1964, Chromosomale Strukturanalyse und Fertilitätsprüfung an einer Margin-alpopulation von *D. subobscura*, *Z. Vererbungsl.* 95:73-81.
- Templeton, A. R., 1980, The theory of speciation via the founder principle, *Genetics* 94:1011-1038.
- Wasserman, M., 1954, Cytological studies on the *repleta* group, *Univ. Tex. Publ.* 5422:130-152.
- Wasserman, M., 1962, Cytological studies of the *repleta* group of the genus *Drosophila*. V. The *mulleri* subgroup, *Univ. Tex. Publ.* 6205:85-118.
- Wasserman, M., 1982, Evolution and speciation in selected species groups. The *Drosophila repleta* group, in: *The Genetics and Biology of Drosophila*, Vol. 3b (M. Ashburner, H. L. Carson, and J. N. Thompson, Jr., eds.), pp. 61-139, Academic Press, New York.
- Wasserman, M., and Koepfer, H. R., 1977, Character displacement for sexual isolation between *Drosophila mojavensis* and *Drosophila arizonensis*, *Evolution* 31:812-823.
- Wasserman, M., and Koepfer, H. R., 1980, Does asymmetrical mating preference show the direction of evolution?, *Evolution* 34:1116-1126.
- Wasserman, M., Heller, J. L., and Zombek, J., 1971, Male determined sexual discrimination in the species *Drosophila pegasa*, *Am. Midl. Nat.* 86:231-235.
- Watanabe, T. K., and Kawanishi, M., 1979, Mating preference and the direction of evolution in *Drosophila*, *Science* 205:906-907.
- Watanabe, T. K., and Kawanishi, M., 1981, Asymmetrical mating success and the phylogeny of *Drosophila*, *Zool. Mag.* 90:317-324.
- Watanabe, T. K., and Kawanishi, M., 1983, Stasipatric speciation in *Drosophila*, *Jpn. J. Genet.* 58:269-274.
- White, M. J. D., 1968, Models of speciation, *Science* 159:1065-1070.